Volume III · Chapter 12 — Retrieval-Augmented Generation (RAG) · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-12-context.md` for the AI interaction log.

Prerequisites: Ch 7, Ch 8, Ch 11.

Learning outcomes — the student can: explain RAG architecture; build an ingest→embed→retrieve→generate pipeline; ground answers with citations; evaluate retrieval and answer faithfulness; reduce hallucination.

Labs (hands-on) — 18 h:

#	Lab	h
12a	Build a basic RAG over clinical guidelines (ingest→embed→retrieve→answer)	5
12b	Chunking experiments: compare strategies on retrieval quality	3
12c	Add citations/provenance to answers	3
12d	Evaluate RAG: retrieval recall + faithfulness scoring	4
12e	Add reranking / hybrid	3

#	Lab	h
	search; measure improvement	

Datasets/tools: LangChain/LlamaIndex or custom; embeddings model; Chroma; local LLM. **Assessment:** working RAG system (**60%**); eval report (**20%**); quiz (**20%**). **Key decisions:** chunk size/overlap; embedding model; RAG vs. fine-tune; citation strategy. **References:** plan §13 → *RAG, vector databases & local inference*. **Hours:** Theory **12** + Lab **18** = **30**.